

Danny Rollo

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EDUCATION

Northeastern University

M.S. in Computer Science

B.S. in Computer Science & Mathematics

Boston, MA

Sept 2024 – April 2026

Sept 2021 – May 2025

PUBLICATIONS

Rollo, D. “Lightweight CNN Encoders for Nucleus Segmentation.” *MICCAI 2026*. **Accepted — poster.**

TECHNICAL SKILLS

Languages & Core Libraries: Python, Go, TypeScript/JavaScript, SQL, pandas, NumPy

Deep Learning: PyTorch, Hugging Face timm, CNNs, U-Net, Vision Transformers (Swin), Transformers & attention

Machine Learning: scikit-learn, InterpretML (EBM), gradient boosting, PCA, SVM, collaborative filtering

LLM & Agentic Systems: Claude and OpenAI APIs, MCP servers, multi-agent orchestration, prompt engineering, RAG, SentenceTransformers

Infrastructure: Docker, AWS (S3), CUDA, SLURM job arrays, submitit, MLflow

SELECTED PROJECTS

Lightweight CNN Encoders for Nucleus Segmentation

PyTorch, Hugging Face timm, PanNuke, MoNuSeg

- Designed and ran a controlled encoder ablation isolating encoder choice in U-Net segmentation, holding decoder, augmentation, and optimization schedule fixed across VGG, ResNet, and Swin-T backbones.
- Established that a 34M-parameter VGG U-Net (Dice 0.851 on PanNuke; 0.796 on MoNuSeg) matches a purpose-built medical transformer and is Pareto-dominant on inference cost.

Two-Tower Recommendation with Cold Start Ablation

PyTorch, BPR loss, Yelp Open Dataset

- Found context features alone (day-of-week, distance, restaurant temporal profiles) outperform the full model on cold-start users by 4.2 NDCG@10 points, while content features remain essential for cold restaurants — an asymmetry in feature importance across cold-start regimes.
- Designed a leave-one-out evaluation protocol preventing train/test leakage, applied to the cold-start case where a user’s only signal comes from held-out reviews.

EXPERIENCE

Pitchcut — Founder & CTO

June 2026 – Present

- Built a computer-vision pipeline (tracking, event/goal detection, player re-identification) for grassroots football match footage, validated against SoccerTrack v2 ground truth, with a consent-gated data flywheel for continued training-set collection.
- Optimized the CV pipeline’s CUDA inference path (single-GPU-upload tile cropping for 17-tile SAHI-style detection, fp16 autocast) for a 2.4× throughput gain (245ms → 102ms/frame) over the CPU-tile fp32 baseline, auto-detected across CUDA/MPS/CPU and parity-verified before shipping.

Teradata — AI/ML Product Management Analytics Intern

June 2025 – May 2026

- Authored the business case and led the design of a context-aware AI Notebook Assistant for VantageCloud AI Studio, enabling intelligent code generation, contextual suggestions, and integration with company libraries and documentation.
- Built frontend prototypes and interaction specs for agentic notebook assistant features, translating LLM capability constraints into user-facing product decisions.

PlateMate — Co-Founder & Chief Technology Officer

May 2025 – Present

- Lead ML architecture and engineering for a food recommendation platform: hybrid item-based collaborative filtering plus cross-modal semantic retrieval over restaurant reviews.
- Built a representation-learning retrieval pipeline with 768-dimensional `a11-mpnet-base-v2` embeddings on MongoDB Atlas Vector Search, parsing natural-language dish queries into semantic and structured geo-aware filters.

LineVision Inc. — Software Engineer Co-op

Jan 2024 – June 2024

- Full-stack owner of an internal web application; defined features with stakeholders, expanded APIs and database schema, maintained the application end to end.
- Re-architected a data pipeline for staff scientists, achieving a 1000× speed improvement.

Generate Product Development — Project Lead, Technical Lead, Software Engineer

Jan 2023 – May 2025

- Led 14 engineers and designers to deliver a production mobile application; defined requirements with external stakeholders.

Khoury College of Computer Sciences — Lead Teaching Assistant

2022 – 2025

Fundamentals of Computer Science I & II · Led weekly two-hour laboratory sections of 40+ students across six semesters, held 1-on-1 office hours, and graded assignments.

HONORS & AWARDS

Rookie of the Year, HackBeanpot · Students of the Year Runner-Up, LLS · Eagle Scout, Scouts BSA